



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc. CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ORGANIC CHEMISTRY-II				
Type of Course	MAJOR				
Semester	IV				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Concept of isomerism: Types of isomerism - constitutional isomerism (chain, position and functional) 2. Basic idea about organic addition reactions, substitution and elimination reactions, aromatic substitution reactions etc.				
Course Summary	The concepts of chirality, Optical isomerism, Relative and absolute configuration and racemic mixture and its separation are included in the first module. The course is designed to provide a comprehensive understanding of addition, substitution and elimination reactions of organic chemistry. The practical component of the course helps to acquire skills in organic synthesis and Column chromatographic techniques.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand the concepts of chirality, optical isomerism and relative and absolute configuration.	U	C	Seminar presentation /Assignment
CO2	To provide a comprehensive understanding of addition reactions in organic chemistry, To understand the mechanisms and stereochemistry of addition reactions.	Ap	P	Class test /Quiz /Assignment

CO3	Understanding the mechanism and stereochemical aspects of substitution reaction at sp ³ carbon.	An	P	Seminar Presentation / Instructor created exam
CO4	To provide a comprehensive understanding of elimination reactions.	U	C	Instructor-created exams / Home Assignments
CO5	Examine the mechanisms and factors influencing aromatic substitution reactions.	Ap	P	Assignment /Seminar presentation /Class test
CO6	Execute practical lab techniques in organic synthesis. Acquire skills in conducting column Chromatography for the separation mixtures. Chromatography for the separation mixtures.	Ap	P	Lab work /Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours	Marks
I	Stereochemistry II		13	29
	1	Optical Isomerism: Optical activity – Concept of chirality – Chirality in organic molecules.	2	
	2	Enantiomers, Diastereomers and Meso compounds.	2	
	3	Optical isomerism in glyceraldehyde, lactic acid and tartaric acid.	1	
	4	Relative and absolute configuration - DL system, RS system of nomenclature for acyclic optical isomers with one and two asymmetric carbon (Amino acids ,Tartaric acids)– sequence rules. Erythro and threo representations (basic idea only)	4	
	5	Racemic mixture – Resolution methods (Chemical and biochemical methods)	2	
	6	Enantiomeric excess, Optical purity. Common approaches in asymmetric synthesis. (mention only)	2	
II	Addition reactions		12	27

	7	Addition reactions to carbon-carbon multiple bonds: Origin of reactivity, regioselectivity (Markownikov's and anti-Markownikov's additions) and stereoselectivity of addition reactions	2	
	8	Examples of addition reactions: hydrogenation, halogenation, hydrohalogenation, hydration, oxymercuration-demercuration,	2	
	9	hydroboration-oxidation, epoxidation, dihydroxylation, ozonolysis.	1	
	10	Addition to $C\equiv C$: Mechanism, reactivity, regioselectivity (Markownikov's and anti-Markownikov's additions) and stereoselectivity	3	
	11	Reactions: Complete hydrogenation, Partial hydrogenation, Electrophilic addition of halogens and hydrogen halides, Ozonolysis	2	
	12	Acidity of alkynes – test for terminal alkynes – Oxidation– (Ozonolysis and reaction with alkaline $KMnO_4$). Chemistry of the tests for unsaturation: Bromine water and Baeyer's reagent test.	2	
III	Substitution and Elimination Reactions		10	21
	13	Nucleophilic substitution reactions: Substitution at sp^3 centre (systems: alkyl halides and alcohols)- Origin of reactivity, SN_1 , SN_2 with stereochemical aspects, types of leaving groups (Oxygen-based and halogen-based).	3	
	14	Effects of substrate structure, solvent, nucleophile, and leaving group on Nucleophilic aliphatic substitution reactions.	3	
	15	Elimination reactions: E_1 , E_2 & $E1CB$ mechanisms. formation of alkenes and alkynes; mechanisms (with evidence), reactivity, regioselectivity (Saytzeff/Hofmann), and stereoselectivity;	3	
	16	competition between substitution and elimination reactions. Syn elimination	1	
IV	Aromatic Substitution Reactions		10	21
	17	Aromatic Electrophilic Substitution: Mechanism of nitration, halogenation, sulphonation, Friedel-Crafts alkylation, and acylation	3	

	18	Synthesis of Aspirin. Ring activating and deactivating groups- Orientating effect of common substituents in aromatic electrophilic substitution.	2	
	19	Electrophilic substitution reactions of Phenols (bromination, nitration and sulphonation)	2	
	20	Preparation of phenolphthalein and Fluorescein	1	
	21	Aromatic nucleophilic substitution: Bimolecular displacement mechanism	1	
	22	Elimination-addition (benzyne intermediate) mechanism.	1	
V	PRACTICALS		30	
	I	1. Separation of binary mixture using solvent extraction (strong acid neutral, basic+neutral and weak acid+neutral compound combinations) 2. Bromination of Cinnamic acid (Green method- Bromide -Bromate mixture) 3. Preparation of dibenzal acetone 4. Nitration of acetanilide 5. Reduction of ethyl acetoacetate by yeast and measurement of optical rotation. 6. Drawing structures using software. 7. Visualization of SN2 reaction using software	24	
		Open Ended: 1. Making models of enantiomers and diastereomers	6	

References:

1. R. T. Morrison, R. N. Boyd, Organic Chemistry, Pearson Education, New Delhi.
2. I. L. Finar, Organic Chemistry, Vol. I, Pearson Education, New Delhi.
3. M. K. Jain, S. C. Sharma, Modern Organic Chemistry, Vishal Publishing Company Co.
4. K. S. Tewari, N. K. Vishnoi, S. N. Mehrotra, A Textbook of Organic Chemistry, Vikas Publishing House.
5. P. Y. Bruice, Essential Organic Chemistry, 3rd Edn., Pearson Education, 2015.

6. John McMurry, Organic Chemistry, 5th Edn., Thomson Asia Pvt. Ltd.
7. C. N. Pillai, Organic Chemistry, Universities Press.
8. Vogel's practical organic chemistry.
9. John McMurry, Eric Simanek, Fundamentals of organic chemistry, 6th Edn., Thomson India Edition.
10. Chemistry for Pharmacy Students: General, Organic and Natural Product Chemistry, S D Sarkar and L Nahar, John Wiley and sons, Ltd.

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	-	-	-	-	-	2		2		2		1
CO 2	2	2	-	-	2	-	3		1		2		1
CO 3	2	-		-	-	2	2		1		2		1
CO 4	2	-		1	-	-	3		1		2		1
CO 5	3		-	-	-	-	2		1		2		1
CO 6	-	-	3		-	-	3		3		2	2	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

Quiz / Assignment/ Quiz/ Discussion / Seminar

Midterm Exam

Programming Assignments (20%)

Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/viva/seminar	Practical skill Evaluation	End Semester Examinations
CO 1		✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	